



The Firth Dinosaurs

NEW ZEALAND'S
FIRST EXHIBITION
EXPLORING DINOSAURS
AND THEIR RELATIVES

AUCKLAND MUSEUM

1987



OCEANBRIDGE

DINOSAURS FLOWN BY



front cover: *Tsintaosaurus*, a
 dinosaur from eastern China,
 one of the centrepieces of the
 exhibition.

109897

Reading Room

**Auckland War Memorial
 Museum Library**

Most of the text and illustrations in this catalogue have been taken from "Esso's Return of the Dinosaurs" (1987), the catalogue of the Melbourne showing of this exhibition at the Museum of Victoria. Credits for text, illustrations and production are given in that publication. Auckland Museum is grateful to the Museum of Victoria for permission to use their material in this New Zealand catalogue, which was coordinated by Richard Wolfe. The text "Dinosaurs in New Zealand" was written by Brian Gill. Printed by McCollam Printers Ltd, Auckland.

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 Private Bag, Auckland,
 New Zealand

1987

ISBN 0-908623-29-1

FOREWORD

Dinosaurs have held an endless fascination since their discovery by scientists early last century. Auckland Museum continually receives enquiries about dinosaurs, especially from school children, yet until now there has been no chance in New Zealand for people to see dinosaurs, to marvel at their size, to compare their different forms, to wonder at their disappearance.

We are proud to be able to present such an opportunity to the people of Auckland and New Zealand, in this the first ever exhibition of dinosaurs in New Zealand. We are indeed grateful to the Museum of Victoria in Melbourne, its Council, Director and staff, for their participation in this joint venture, which we hope will be the first of many.

Firth Concrete Products have willingly undertaken this major sponsorship, and to them, their staff and their consultants I express the Museum's thanks. Qantas and Oceanbridge have been of great assistance in transporting both dinosaurs and dinosaur experts from Melbourne, and Gray Bartlett Enterprises were also of assistance in the early stages of the exhibition planning.

There is an Australian emphasis to the details in this catalogue and to the exhibition itself. We make no apology for this. Until recently dinosaurs were thought to have been absent from New Zealand. This is why our museums do not display them prominently. Any big dinosaur display will consist largely of material not directly relevant to New Zealand. However, we have been careful to assemble a small selection of local specimens to give the exhibition a New Zealand perspective.

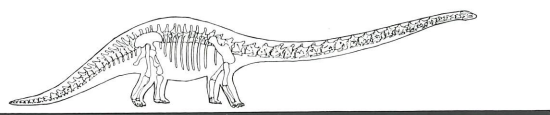
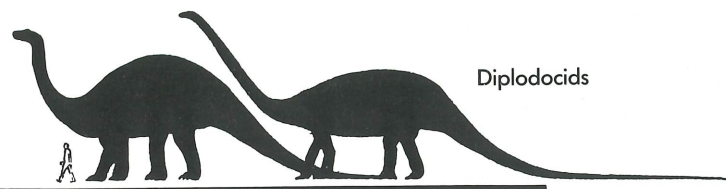
Many Auckland Museum staff have helped to make the exhibition and this catalogue a reality. We have also been helped by other specialists within New Zealand, and for their ready cooperation we are also grateful. The opportunity to display New Zealand's first recorded dinosaur is particularly appreciated — the cast was prepared by Ian Keyes of the New Zealand Geological Survey, Lower Hutt.

We are sure many thousands of visitors to Auckland Museum will enjoy this exhibition, will learn from it, and will marvel at these wondrous animals. It is our pleasure to present "The Firth Dinosaurs".

Stuart Park
Director, Auckland Institute and Museum.

SAUROPODS

DIPLODOCIDS



Diplodocids are the most often thought of dinosaurs – large, quadrupedal (standing on four feet), long tailed, long necked reptiles. Generally, they had relatively small brains and were restricted to the Late Jurassic and Cretaceous (155-65 m. yrs ago). They had peculiar skid-like bones on the whiplash-like tail, and rather short front legs. Their feet appear to have been elephant-like in having a wedge of fibrous tissue lying beneath the rear of the foot, rather like the heel of a shoe, that supported the toe bones in an upright stance.

Mamenchisaurus (ma-men-key-sawr-us)

Meaning:

'Mamenchisaurus reptile.' Mamenchis is the locality in the People's Republic of China where this dinosaur was found.

Age:

Late Jurassic (140-155 m. yrs)

Locality:

China

Diet:

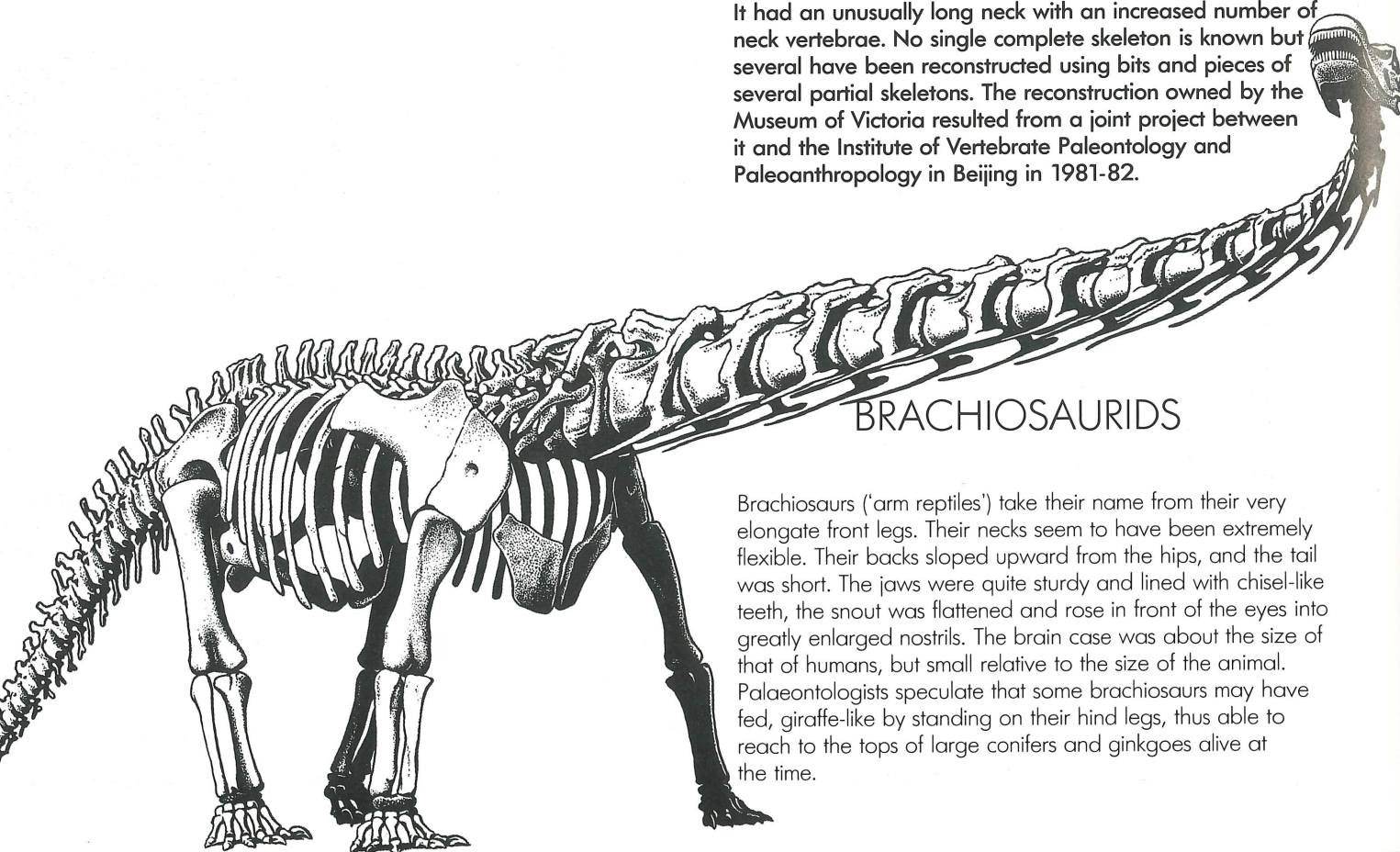
Plant eater

Size:

22 metres long, 3.5 metres high at the shoulder.

Comments:

Mamenchisaurus was a very large sauropod that weighed up to 40-50 tonnes and may have been related to diplodocids. It had an unusually long neck with an increased number of neck vertebrae. No single complete skeleton is known but several have been reconstructed using bits and pieces of several partial skeletons. The reconstruction owned by the Museum of Victoria resulted from a joint project between it and the Institute of Vertebrate Paleontology and Paleoanthropology in Beijing in 1981-82.



BRACHIOSAURIDS

Brachiosaurs ('arm reptiles') take their name from their very elongate front legs. Their necks seem to have been extremely flexible. Their backs sloped upward from the hips, and the tail was short. The jaws were quite sturdy and lined with chisel-like teeth, the snout was flattened and rose in front of the eyes into greatly enlarged nostrils. The brain case was about the size of that of humans, but small relative to the size of the animal. Palaeontologists speculate that some brachiosaurs may have fed, giraffe-like by standing on their hind legs, thus able to reach to the tops of large conifers and ginkgoes alive at the time.

Ultrasaurus (ultra-sawr-us)

Meaning:

'Very large reptile'

Age:

Late Jurassic (140-155 m. yrs)

Locality:

Western Colorado, U.S.A.

Diet:

Plant eater

Size:

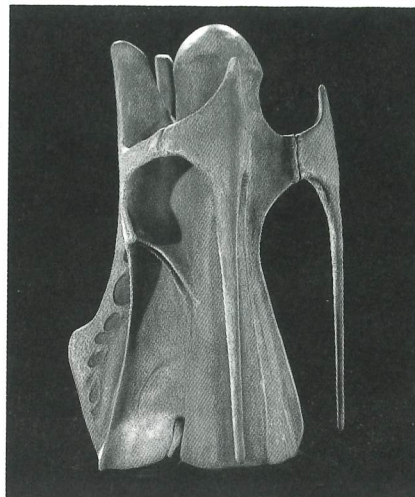
More than 30 metres long.

Exhibit:

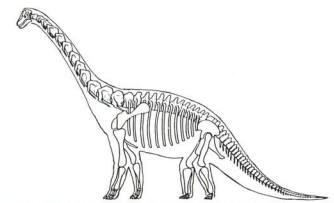
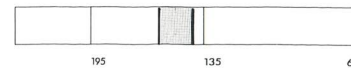
Shoulder blade (scapula) and neck vertebra found by Dr. Jim Jensen of Brigham Young University, Provo, Utah.

Comments:

Ultrasaurus was a brachiosaur and is believed to be one of the largest dinosaurs ever discovered, weighing more than 130 tonnes.

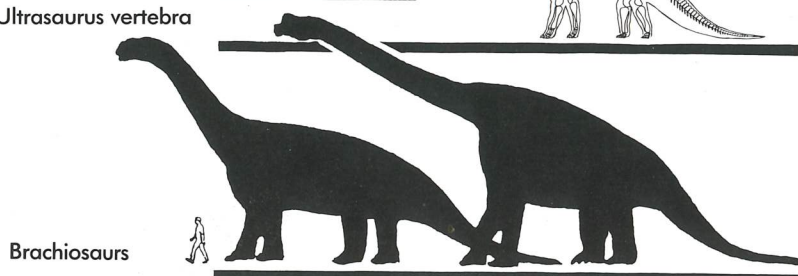


Ultrasaurus vertebra



CETIOSAURS

The 'giant reptiles' were a group of dinosaurs, mainly of Jurassic age that had relatively short necks, and distinctive vertebrae. They may all be related to one another or a collection of forms that are not actually related. They lacked cavities, or pleurocoels, in the neck and back vertebrae. In dinosaurs that possessed pleurocoels, it was a way of lightening up the heavy bones. They also lacked the groove or cleft in the vertebral spine which held a strong, energy saving ligament that attached to the base of the skull. Some of these animals must have been truly terrestrial, while others frequented water. Trackways of these and other sauropods show that some moved through water with only the front feet touching. Some also travelled in herds like modern elephants and perhaps protected their young by encircling them.



Brachiosaurs

Rhoetosaurus (row-stow-sawr-us)

Meaning:

'Rhoetos' was a giant in mythology. 'Giant reptile'

Age:

Middle Jurassic (155-170 m. yrs)

Locality:

Southeastern Queensland

Diet:

Plant eater

Size:

12.5 metres long (estimated)

Exhibit:

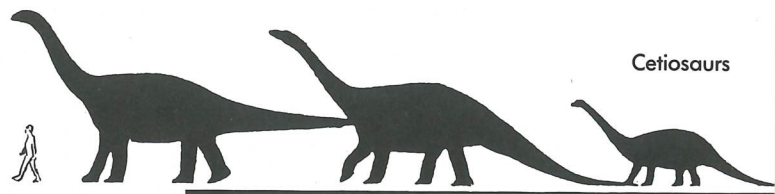
Thigh-bone discovered near Roma, Queensland, in 1924 by A. J. Browne in association with the Queensland Museum

Comments:

Rhoetosaurus brownei is one of the earliest sauropods known. It is based mainly on the hind legs and tail from the Injune Creek beds and currently being studied by Dr. R. A. Thulborn at the University of Queensland.



Rhoetosaurus bone



Cetiosaurs

